JAVA SINGLE INHERITANCE PROGRAM

Object-Oriented Programming Practical Record

1. Aim

To write a Java program using **single inheritance** to implement a calculator that performs addition, subtraction, multiplication and division.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named Main.java.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create an object for AdvancedCalculator.
6. AdvancedCalculator inherits the properties and methods of Calculator.
7. Call the addition, subtraction, multiplication and division methods.
8. Display the results of all calculator operations.

3. Algorithm

1. Start the program.
2. Create the parent class Calculator.
3. Declare two integer variables a and b.
4. Create addition() and subtraction() methods in the Calculator class.
5. Create the child class AdvancedCalculator extending Calculator.
6. Create multiplication() and division() methods in the child class.
7. Create an object of AdvancedCalculator in main().
8. Call all four calculator methods.
9. Display the results.
10. Stop the program.

4. Explanation of the Program

Calculator class: It is the parent class containing two integer variables, a and b. It also contains the addition() and subtraction() methods.

AdvancedCalculator class: It is the child class that extends the Calculator class. It inherits the variables and methods of the parent class and provides multiplication() and division() methods.

Single Inheritance: The AdvancedCalculator class inherits from only one parent class, Calculator. Therefore, the program demonstrates single inheritance.

Main class: The main() method creates an object of AdvancedCalculator and calls all four calculator operations.

5. UML Class Diagram

```
+--------------------------------+
|          Calculator            |
+--------------------------------+
```

```
| - a : int = 20                  |
| - b : int = 10                  |
+---------------------------------+
| + addition() : void             |
| + subtraction() : void          |
+----------------+----------------+
                 |
                 | extends
                 v
+---------------------------------+
|      AdvancedCalculator         |
+---------------------------------+
| + multiplication() : void       |
| + division() : void             |
+---------------------------------+

                 |
                 v
+---------------------------------+
|             Main                |
+---------------------------------+
| + main() : void                 |
+---------------------------------+
```

6. Java Source Code

```
// Single Inheritance
// Real application: Calculator

class Calculator {
    int a = 20;
    int b = 10;

    void addition() {
        System.out.println("Addition = " + (a + b));
    }

    void subtraction() {
        System.out.println("Subtraction = " + (a - b));
    }
}

class AdvancedCalculator extends Calculator {

    void multiplication() {
        System.out.println("Multiplication = " + (a * b));
    }

    void division() {
        System.out.println("Division = " + (a / b));
    }
}

public class Main {
    public static void main(String[] args) {

        AdvancedCalculator obj = new AdvancedCalculator();

        obj.addition();
        obj.subtraction();
        obj.multiplication();
        obj.division();
    }
}
```

7. Output of the Program

```
Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2
```

8. Result

Thus, the Java program successfully demonstrates **single inheritance** using a real-world calculator application. The AdvancedCalculator class inherits the addition and subtraction operations from the Calculator class and performs multiplication and division operations.